Adaptive Path Planning for AUVs in Dynamic Aquaculture via DDPG-GAN Synergy

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Abstract. Autonomous underwater vehicles operating in dynamic aquaculture environments face significant challenges from unpredictable obstacles including fish swarms and ocean currents, demanding real-time adaptive path planning to ensure both navigational efficiency and operational safety. While existing algorithms exhibit limitations in such conditions, this work proposes an adaptive path planning framework integrating deep deterministic policy gradient with generative adversarial networks. The framework specifically addresses dynamic obstacle avoidance under stringent real-time constraints. The generative adversarial network component employs adversarial training to simulate complex hydrodynamic interactions, enabling predictive modeling of environmental dynamics. Concurrently, the deep deterministic policy gradient agent optimizes velocity adjustments based on these environmental predictions, facilitating proactive trajectory planning. A hybrid reward mechanism balances critical objectives encompassing collision avoidance, energy conservation, and target convergence through continuous real-time feedback. Extensive simulation within representative aquaculture settings validates the framework, demonstrating substantial performance enhancements. Compared to baseline deep deterministic policy gradient under dynamic conditions, the framework achieves a 22.7% higher mission success rate, a 27.4% improvement in path optimality, and a 37.0% increase in dynamic obstacle avoidance effectiveness. Furthermore, it surpasses traditional algorithms including RRT* and A* in path efficiency and collision-free operation rates, confirming its robustness, real-time adaptability, scalability, and autonomy within continuously disturbed aquaculture ecosystems.

Keywords: FAdversarial Learning · AUVs · Collision Avoidance · DDPG · Dynamic Aquaculture Modeling · Energy Efficiency, GANs · Hybrid Rewards · Hydrodynamic Models · Obstacle Strategies · Path Planning · Real-Time Adaptation.

1 Introduction

Marine aquaculture, an intensive form of aquaculture conducted in oceanic or coastal environments, typically employs floating or fixed cages for the cultivation of fish and other marine organisms. To monitor and enhance aquaculture conditions, it is imperative to collect data on environmental parameters such as water quality, temperature, dissolved oxygen levels, and pH values. This information is critical for ensuring the health and productivity of farmed species. Automated aquaculture involves the application of modern scientific and technological methods, automated equipment, and intelligent systems to manage and optimize all facets of aquaculture operations.

Currently, there is limited research on robot motion planning within autonomous aquaculture systems. Examples include motion planning for autonomous underwater vehicles (AUVs) used in oyster farm maintenance, cage cleaning, and mesh networking applications. In 2024, Wang et al. introduced an overpass obstacle avoidance search algorithm designed for Autonomous Underwater Vehicles (AUVs) that adapts to ocean currents; however, this approach considered only the single factor of ocean currents.

Moreover, traditional methods generally yield satisfactory results in scenarios with fixed obstacles or explorations involving a single influencing factor[1]. However, they are relatively inadequate for path planning in environments where multiple interacting factors (such as fish behavior) are present.

The increasing demand for aquaculture products, coupled with labor shortages and limited existing research, has made this topic a priority for further investigation. In this context, underwater drones play a crucial role in aquaculture by facilitating navigation, secure communications, reconnaissance, parcel delivery, precision agriculture and fisheries, and search and rescue operations.

Research into autonomous aquaculture systems has become increasingly common [2], with drones already being utilized for various applications in aquaculture management, including water quality monitoring, biomass estimation, underwater inspection, and intelligent feeding. These applications highlight the potential of drones to enhance aquaculture efficiency and mitigate risks. However, these technologies face limitations such as environmental adaptability, data precision, economic viability and system integration challenges, which collectively hinder their widespread adoption and consistent performance in real-world aquaculture settings.

In this paper, an adaptive motion planning framework is proposed to enhance the motion planning capability of AUVs in aquaculture environments. In addition to effectively avoiding traditional static reef obstacles, the advantage of our path planning method is that it can effectively deal with complex dynamic scenes. This addresses the challenges posed by dynamic changes in the aquaculture environment, such as school movement and ocean currents. This paper innovatively trains the Generative Adversarial Networks(GAN) network through Deep Deterministic Policy Gradient(DDPG), which greatly improves the autonomous navigation ability of AUV compared with the traditional path planning algorithm, and meets the requirements of automatic AUV navigation under complex water conditions in real scenes.

The contributions of our work are as follows:

– A novel adaptive motion planning framework integrating DDPG and GANs, specifically designed for dynamic aquaculture environments. This framework

enables AUVs to simultaneously address static obstacles (e.g., reefs) and dynamic factors (e.g., fish swarms and ocean currents).
- The first implementation of GANs trained via DDPG for underwater navigation, where the GAN dynamically simulates complex environmental conditions whereas the DDPG agent optimizes real-time path planning through adversarial learning.
- A multi-dimensional state space formulation incorporating both kinematic parameters (position/velocity) and environmental dynamics (current velocity, fish density), enabling comprehensive environment perception for AUVs.
- A hybrid reward mechanism is proposed, which includes three original components: directional guidance through distance-proportional arrival reward, collision avoidance via obstacle radius-based penalty, and energy efficiency optimization using velocity-norm penalty.
- Extensive validation performed in simulated aquaculture environments highlights substantial performance improvements. Specifically, the DDPG+GAN framework achieves enhancements in success rate, path optimality, and dynamic obstacle avoidance effectiveness, with respective gains of 22.7%, 27.4%, and 37.0% compared to the baseline DDPG algorithm under dynamic conditions. Furthermore, under dynamic conditions, the proposed algorithm exhibits significantly improved path efficiency and a considerably higher non-collision rate in comparison to traditional algorithms such as RRT* and A* [1].

2 Related Work

Contemporary path planning methodologies for autonomous underwater vehicles (AUVs) primarily focus on heuristic-based algorithms, reinforcement learning architectures, generative adversarial network integrations, and model-driven frameworks. These methodologies collectively tackle the challenges of navigating dynamic obstacles and adapting to environmental changes whereas addressing limitations in computational efficiency and real-time responsiveness under uncertain conditions.

Recent advancements in path planning for autonomous underwater vehicles (AUVs) and unmanned aerial vehicles (UAVs) have successfully combined classical algorithms with contemporary machine learning techniques [3]. Traditional methods, such as the enhanced A^* algorithm specifically designed for marine robots, emphasize heuristic adaptations to oceanic environments but face challenges in handling dynamic obstacles and ensuring real-time adaptability [1]. Evolutionary algorithms, particularly those optimized for variable ocean currents, utilize dynamic programming to enhance robustness, yet they encounter computational inefficiencies in high-dimensional state spaces [4].

The integration of Reinforcement Learning (RL) has substantially advanced autonomous navigation capabilities. Multi-layer Q-learning frameworks, as introduced to break down path planning into hierarchical tasks to effectively balance exploration and exploitation in complex environments [5]. Similarly, RL techniques have been successfully applied to edge computing networks, optimizing

UAV trajectories through Gauss-Markov stochastic models to minimize latency [6]. For multi-AUV systems, adaptive formation control using actor-critic networks exhibits improved collision avoidance and cooperative behavior in cluttered underwater environments [7].

To further enhance the performance of reinforcement learning-based planning algorithms in dynamic and uncertain environments, recent studies have highlighted the importance of integrating generative adversarial networks (GANs) for synthesizing high-fidelity training data and optimizing decision-making processes across a wide range of robotic applications. GANs have emerged as powerful tools for enhancing RL-based planning. Li et al. (2024) leverage conditional GANs to generate heuristic samples in narrow spaces, thereby improving computational efficiency for robotic manipulators [8]. Chu (2024) further integrates GANs with DRL to enhance exploration efficiency, effectively mitigating sample complexity in sparse reward scenarios [9]. In UAV navigation, GAN-based path planners optimize reconfigurable intelligent surface-assisted communication, demonstrating robust performance in urban areas with dynamic obstacles [10]. Recent work by Lee et al. (2025) combines GANs with hindsight experience replay (HER) to bridge the sim-to-real gap, achieving data-efficient training for quadcopter control [11].

Model-based approaches, such as APPA-3D, implement autonomous 3D path planning in unknown environments through adaptive risk assessment and hierarchical optimization, although they remain heavily dependent on static environmental assumptions [12]. PlanGAN addresses sparse rewards by synthesizing plausible trajectories but lacks real-time adaptability in continuously varying conditions [13]. In contrast, this study advances physics-aware environmental modeling through adversarial training, enabling precise prediction of oceanic disturbances (e.g., currents, fish swarms) and facilitating real-time path replanning. By integrating high-fidelity simulations with depth estimation techniques, the framework outperforms model-based benchmarks in terms of collision avoidance efficiency and operational stability, as validated on AirSim's 3D platform [14]. These innovations address critical limitations in existing literature, particularly in dynamic aquaculture ecosystems where energy efficiency and real-time responsiveness are essential.

3 Prior knowledge

3.1 Deep reinforcement learning

Deep reinforcement learning (DRL) has emerged as a formidable approach for controlling and decision-making processes, particularly in high-dimensional spaces [7]. By fusing reinforcement learning (RL) with deep neural networks, DRL is capable of learning intricate mappings from raw sensory inputs to actions. In this discussion, we will focus in particular on the advantages of the Depth Deterministic Strategy Gradient (DDPG) approach.

Reinforcement Learning (RL) operates on the principle of learning by interaction with an environment. It involves an agent that selects actions in an attempt to maximize some notion of cumulative reward. The agent's performance is guided by the Markov Decision Process (MDP) framework, which models the environment as a set of states, actions, transition probabilities, and rewards. The MDP provides a structured approach to decision-making under uncertainty, where the agent's goal is to find a policy that maps states to actions to achieve the optimal cumulative reward.

The Depth Deterministic Policy Gradient (DDPG) method is a type of deep reinforcement learning, which addresses the challenges of continuous action spaces by combining the actor-critic framework with experience replay and target networks. The critic estimates the action-value function, whereas the actor learns to select actions that maximize this value:

$$Q(s, a \mid \theta^Q) = r + \gamma Q\left(s', \mu(s' \mid \theta^\mu) \mid \theta^Q\right) \tag{1}$$

$$\nabla_{\theta^\mu} J \approx \mathbb{E}\left[\nabla_a Q(s, a \mid \theta^Q)\big|_{a=\mu(s)} \cdot \nabla_{\theta^\mu} \mu(s \mid \theta^\mu)\right] \tag{2}$$

DDPG's ability to handle continuous actions and its robustness in high-dimensional spaces make it particularly advantageous for UAV control tasks that demand fine-grained, smooth, and responsive actions [15].

3.2 GAN

Generative Adversarial Networks (GANs) are a machine learning framework designed to generate synthetic data that closely mimics real data. GANs consist of two neural networks:

Generator$G(z|\theta_G)$ generates synthetic data from random noise z as input, with the objective of producing outputs that are indistinguishable from real data [9]. In this study, the generator predicts environmental conditions such as ocean currents and fish density based on drone positioning data.

Discriminator$D(x|\theta_D)$ assesses the probability that the input x is real. It aims to maximize correct classification, whereas the generator seeks to minimize the discriminator's ability to distinguish between real and synthetic data. Specifically, the discriminator differentiates between actual environmental observations and generated data. Adversarial training is formalized as follows:

$$\begin{aligned}\min_G \max_D V(D, G) = & \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] \\ & + \mathbb{E}_{z \sim p_z(z)} [\log (1 - D(G(z)))]\end{aligned} \tag{3}$$

In this paper, GANs simulate dynamic environmental factors, which are crucial for drone navigation in aquaculture [10]. When integrated with DDPG, the system can better predict and adapt to underwater changes, thereby enhancing UAV performance in dynamic and uncertain environments [8].

3.3 AUV basics

Unmanned Aerial Vehicles (UAVs), also known as drones, are aircraft systems without a human pilot aboard. They are controlled either autonomously by onboard computers or remotely by operators on the ground. UAVs are used for various purposes such as surveillance, delivery, data collection, and more. UAVs typically consist of an airframe, propulsion system, flight control system, sensors, and a communication system. We will use the classic quadcopter UAV provided by AirSim to simulate the AUV described below, as shown in Fig. 1.

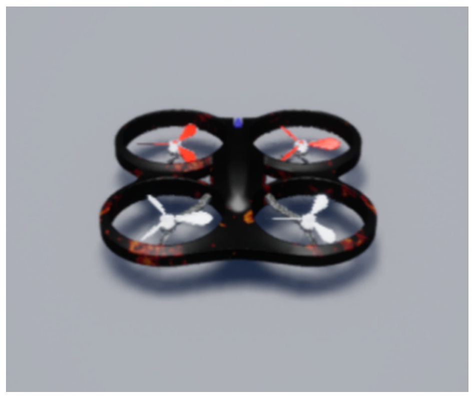

Fig. 1. UAV model in airsim

Autonomous Underwater Vehicles (AUVs) are unmanned robotic systems specifically engineered to execute mission-specific operations in subsea environments over extended durations, ranging from hours to days. These systems depend on sophisticated control and navigation architectures to ensure the effective execution of their tasks. AUVs play a pivotal role in various critical applications, such as seabed mapping, geological hazard assessment, hydrocarbon infrastructure inspection, biological parameter monitoring, and hydrodynamic research.

Unlike terrestrial or aerial drones, AUVs operate in inherently complex and unpredictable underwater domains marked by dynamic hydrographic conditions, unstructured obstacle distributions, limited visibility, and constrained communication bandwidth. To guarantee operational integrity and mission success in these challenging environments, robust real-time path planning algorithms and adaptive dynamic obstacle avoidance systems are indispensable [16]. Optimal trajectory generation must account for spatially varying currents [17], transient obstacles, and terrain topology whereas minimizing energy consumption [4]. Furthermore, dynamic obstacle avoidance capabilities are vital for reducing collision risks with marine fauna, submerged structures, or unforeseen geological features, particularly in poorly charted regions [18]. The integration of multi-sensor fusion, predictive modeling, and reactive control frameworks significantly enhances the vehicle's autonomy, enabling reliable navigation in cluttered or rapidly changing subsea environments where traditional preprogrammed paths may prove inadequate [19].

Table 1. Notations

Symbol	Description
$P = (x, y, z)$	AUV's position
(v_x, v_y, v_z)	AUV's velocity
v_{cx}, v_{cy}	current velocity on the X-axis, Y-axis
ρ	fish density at the current position
(x_f, y_f, z_f)	position of fish swarm center
d	distance between AUV and fish swarm
t	current time
$(\Delta v_x, \Delta v_y, \Delta v_z)$	AUV's velocity variations
d_{target}	distance between AUV and target
$\mathcal{R}_{arrival}$	arrival reward
$\mathcal{R}_{total}$	total reward
$\mathcal{P}_{collision}$	collision penalty
$\mathcal{P}_{energy}$	energy penalty
p_{obs}	obstacles' position
r_{obs}	distance between AUV and obstacles
r_{obs}	AUV's velocity vector
L_a	Algorithm generates path length
L_t	Theoretical optimal path length
$v_{current}$	Velocity of ocean current
v_{base}	Base flow rate:[0.2,0.1]m/s
A	Velocity fluctuation amplitude:0.1m/s
T	Period:60s

4 Methodology

In this section, we present a novel framework that significantly enhances the autonomous navigation capabilities of autonomous underwater vehicles (AUVs) compared to traditional path planning algorithms. This advanced approach is specifically designed to meet the stringent requirements for automated navigation in complex aquatic environments. The framework adopts a hierarchical learning architecture [5], integrating a generative adversarial network (GAN) for environmental modeling with a Deep Deterministic Policy Gradient (DDPG) algorithm for strategy learning. At the higher level, the GAN predicts environmental dynamics such as currents and obstacles, whereas at the lower level, the DDPG optimizes immediate control actions, thereby enabling efficient adaptation to dynamic underwater conditions. This sophisticated synergy between reinforcement learning and generative techniques not only improves path planning but also facilitates more robust and intelligent decision-making processes.

The problem is formalized using Markov Decision Processes (MDP), wherein the state space, action space, and reward function are rigorously defined to encapsulate the intricacies of underwater navigation [20]. Consequently, this ensures that AUVs can autonomously adapt to challenging and unpredictable scenarios.

The implementation of the DDPG algorithm enables continuous and precise control over the AUV's navigation in highly dynamic and uncertain environments, thereby facilitating efficient exploration of complex state spaces. In the existing reinforcement learning framework, we incorporate the temporal difference (TD) update mechanism from Q-learning to refine the estimation of the value function. Furthermore, by combining this approach with a prioritized sampling strategy from the experience replay buffer, we achieve substantial improvements in the agent's exploration efficiency within sparse reward environments [6]. Furthermore, the algorithm's capacity to learn from historical data through a deep reinforcement learning framework significantly enhances the vehicle's decision-making capabilities, thereby ensuring optimal performance across diverse underwater conditions [21].

4.1 RL-based Approach

State Space In the DDPG algorithm, the action space of AUV is continuous, allowing it to take any value within a certain range of actions. The environment considered in this study is three-dimensional. Therefore, we describe the position of the AUV in the environment using coordinate points and take the position as part of the state variable S of the MDP model [7]. The state space for the AUV is defined as follows:

Position (3 dimensions): x, y, and z indicate the AUV's position on the X-axis, Y-axis, and Z-axis respectively.

Velocity (3 dimensions): v_x, v_y and v_z indicate the AUV's velocity on the X-axis, Y-axis, and Z-axis respectively.

Environmental Parameters (3 dimensions): v_{cx} and v_{cy} indicate the current velocity on the X-axis and Y-axis respectively [22] [23]. In addition to this, ρ denotes the fish density at the current position.

Thus, the AUV's state space can be represented as:

$$S = [x, y, z, v_x, v_y, v_z, v_{cx}, v_{cy}, \rho] \tag{4}$$

In addition to that, the state space for dynamic obstacles (fish swarm) is not explicitly defined as a separate state vector in the code but is dynamically calculated using the Fish density formula [24]. The state of the fish swarm can be understood as:

Fish Swarm Center Position (3 dimensions): x_f, y_f, and z_f indicate the position of fish swarm center on the X-axis, Y-axis, and Z-axis respectively.

Fish Density (1 dimension): The fish density calculated based on the distance between the AUV and the fish swarm center.

The position of the shoal changes dynamically. We assume the periodic movement of the shoal around the point $(50, 50, -10)$, and let the shoal center point be (x_f, y_f, z_f), whose x and y coordinates swing periodically with time, similar to the trajectory of fish swimming in water. Such motion trajectories can often be modeled by sinusoidal functions. Therefore, the following calculation formula of position change with time can be obtained:

$$\mathbf{f_c(t)} = \begin{pmatrix} x_f + 10\sin\left(\frac{t}{10}\right) \\ y_f + 10\cos\left(\frac{t}{10}\right) \\ -10 \end{pmatrix} \tag{5}$$

At the same time, in order to enhance the simulation effect, we dynamically calculate the fish density according to the distance between the AUV and the fish center. Setting AUV's position $P = (x_p, y_p, z_p)$ and fish_center $f_c(t) = (x_f, y_f, z_f)$, the distance computes the Euclidean distance between two points:

$$\begin{aligned} d &= |P - f_c(t)| \\ &= \sqrt{(x_p - x_f)^2 + (y_p - y_f)^2 + (z_p - z_f)^2} \end{aligned} \tag{6}$$

We calculate fish_density from distance, making sure that its value is between 0 and 1, and that the density decreases with distance:

$$\rho = \max\left(0, 1 - \frac{d}{20}\right) \tag{7}$$

Thus, the AUV's state space can be represented as:

$$S = [x_f, y_f, z_f, \rho] \tag{8}$$

Action Space The objective of the AUV is to develop a control strategy by adjusting the propeller thrust and rudder angle to maintain a specified distance from obstacles. To achieve this, the action space is defined as a three-dimensional continuous space denoted by:

$$S = [\Delta v_x, \Delta v_y, \Delta v_z] \tag{9}$$

which representing the velocity variations in the x, y, and z directions used to control the AUV's motion. Considering the dynamics and environmental constraints of the AUV and the mission requirements for smooth motion, precise control, and energy efficiency, the speed in each dimension is constrained to the range $[-5.0, 5.0]$.

Reward Function In this study, the design of the reward function is a core component of the reinforcement learning algorithm. To ensure that the UAV can efficiently and safely complete tasks in complex aquaculture environments, we have designed a multi-faceted reward mechanism. Specifically, the reward function consists of the following components: Arrival Reward, Collision Penalty, Energy Penalty, and Total Reward. The settings for each component are described below:

The Arrival Reward is designed to encourage the AUV to approach the target position as quickly as possible. We define the Arrival Reward by calculating the Euclidean distance between the UAV's current position and the target position. The closer the UAV is to the target, the higher the reward. The specific formula is as follows:

$$\mathcal{R}_{arrival} = -0.1 \cdot d_{target} \tag{10}$$

where d_{target} is the distance between the AUV's current position and the target position. By multiplying by a negative coefficient (0.1), we ensure that the AUV receives a positive reward as it approaches the target, thereby incentivizing it to move toward the goal.

The Collision Penalty is used to prevent the AUV from colliding with obstacles. We determine whether a collision has occurred by measuring the distance between the AUV and all obstacles. If the distance between the AUV and any obstacle is less than the obstacle's radius, a collision is considered to have occurred, and a large negative reward (penalty) is applied. The specific formula is as follows:

$$\begin{aligned} \mathcal{P}_{collision} = -10 \cdot \sum_{\text{obs}\in\text{obstacles}} \\ \mathbb{I}\left(\|p - p_{obs}\|\right. \\ \left. < r_{obs}\right) \end{aligned} \tag{11}$$

where dist is the distance between the AUV and the obstacle, and R_{obs} is the radius of the obstacle. By applying a large negative reward, we ensure that the AUV avoids collisions with obstacles during training.

To further enhance the performance of the AUV, we have conducted a thorough analysis of various factors to develop an energy penalty mechanism that promotes energy-efficient operation. The Energy Penalty is specifically designed to quantify the relationship between the AUV's velocity and its energy consumption. By calculating the norm of the AUV's velocity vector, we can more accurately estimate energy consumption, recognizing that higher velocities lead to increased energy expenditure and, consequently, a greater penalty. This approach systematically addresses the challenge of optimizing energy efficiency during AUV operations. The specific formula used for calculating the energy penalty is as follows:

$$\mathcal{P}_{energy} = -0.01 \times \|\mathbf{v}\| \tag{12}$$

where $||\mathbf{v}||$ is the norm of the AUV's velocity vector. By applying a small negative coefficient (0.01), we ensure that the AUV minimizes unnecessary energy consumption during flight.

subsectionTotal Reward The Total Reward is the weighted sum of the above three components, used to comprehensively evaluate the AUV's performance at each step. The specific formula is as shown in Eq (13).

$$\mathcal{R}_{total} = \mathcal{R}_{arrival} + \mathcal{P}_{collision} + \mathcal{P}_{energy} \tag{13}$$

By combining the Arrival Reward, Collision Penalty, and Energy Penalty, we obtain a comprehensive reward value that guides the AUV's decision-making in complex environments. The design of the Total Reward aims to balance the AUV's task completion efficiency, safety, and energy consumption, ensuring its effectiveness and reliability in practical applications.

4.2 GAN Component Architecture

The Generative Adversarial Network (GAN) framework comprises two key neural networks: a generator and a discriminator, which are specifically designed to model dynamic environmental perturbations in aquaculture ecosystems. The generator synthesizes high-fidelity environmental parameters, such as ocean current velocities and fish swarm density distributions, based on real-time positional data collected by the Autonomous Underwater Vehicle (AUV). It serves as a predictive engine, transforming spatial coordinates into multi-dimensional environmental states through fully connected layers with leaky ReLU and hyperbolic tangent activations. In contrast, the discriminator assesses the authenticity of environmental observations by distinguishing between actual sensor measurements and synthetic outputs generated by the generator. Through adversarial training, the generator progressively refines its predictions to accurately replicate real-world environmental dynamics, whereas the discriminator enhances its capacity to identify discrepancies, thereby establishing a robust feedback loop for precise environmental simulation.

4.3 GAN-DDPG Synergy Mechanism

The integration of GAN with Deep Deterministic Policy Gradient (DDPG) is facilitated through a hierarchical learning architecture, where the generator's predictions enrich the AUV's state space, as shown in Fig. 2. During policy optimization, the DDPG agent leverages the generator's synthesized environmental parameters—such as current velocities and fish density—to construct a comprehensive state representation, enabling proactive trajectory adjustments in dynamic scenarios. The discriminator's adversarial feedback is utilized to fine-tune the generator's output, ensuring consistency with real-world environmental dynamics. Concurrently, the DDPG critic network evaluates state-action pairs using these enhanced states, whereas the actor network generates velocity adjustments that account for both immediate obstacles and anticipated environmental changes. This bidirectional interaction empowers the AUV to anticipate and adapt to dynamic disturbances, such as transient fish swarms and variable currents, thereby optimizing collision avoidance, energy efficiency, and maintaining real-time responsiveness.

5 Experiments

This section focuses on validating the performance and operational feasibility of the proposed DDPG-GAN synergy framework for adaptive path planning of Au-

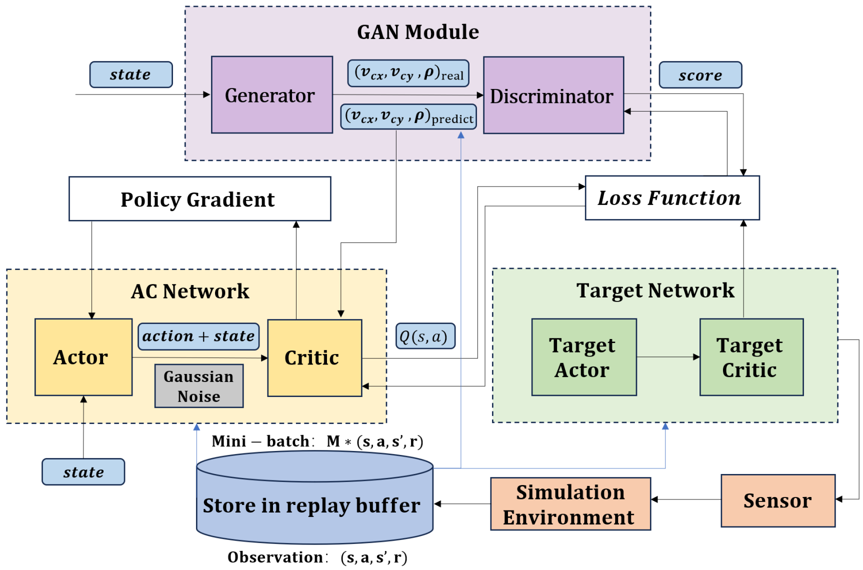

Fig. 2. Framework of GAN+DDPG algorithm

tonomous Underwater Vehicles (AUVs) in dynamic aquaculture environments. Systematic experiments are designed to address the following critical **R**esearch **Q**uestions **(RQ)**:

– **RQ1:** To what extent can the DDPG-GAN synergy mechanism ensure collision-free trajectory generation with high mission success rates under complex aquaculture constraints, whereas simultaneously balancing obstacle avoidance and path reliability?
– **RQ2:** How effectively does the algorithm minimize both path length and energy consumption whereas maintaining real-time responsiveness in large-scale aquaculture scenarios that involve continuous environmental updates?
– **RQ3:** To what extent can the system effectively manage complex dynamic environments, including varying-speed ocean currents and unpredictable fish swarms, whereas ensuring trajectory smoothness and navigation stability under real-time constraints?

Rigorous quantitative evaluations are conducted, including trajectory optimality analysis and computational complexity benchmarking, to compare the proposed framework with traditional optimization methods such as DDPG. The dynamic obstacle avoidance capabilities of the system are further validated through multi-scenario simulations that incorporate stochastic environmental disturbances.

5.1 Experiment settings

In the experiment, we employ two algorithms as our targeted object of study: DDPG and DDPG+GAN. There are relatively few methods comparing these two algorithms in the field of deep reinforcement learning, and we choose DDPG as our baseline for comparison.

Simulation. The simulation framework is constructed using AirSim 1.8.1 integrated with Unreal Engine 4.27.2, enabling high-fidelity underwater scene rendering with photorealistic dynamic ocean currents and biologically inspired fish swarm behaviors. AirSim's physics engine, in conjunction with Unreal Engine's particle-based fluid dynamics, ensures millimeter-level accuracy in depth estimation, which is essential for validating autonomous obstacle avoidance in complex marine environments [3]. The platform leverages AirSim's low-level flight controllers to stabilize AUVs whereas synchronizing with Visual Studio 2022 via APIs for real-time data exchange and efficient training pipelines. This integration effectively bridges the simulation-to-reality gap by utilizing ground-truth depth maps for precise environmental perception [14].

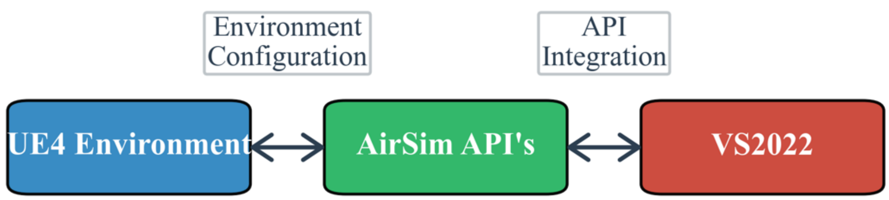

Fig. 3. Obstacle avoidance performance as a function of training cycles

Environmental Perception The proposed framework employs multimodal data to achieve precise environmental perception and dynamic obstacle avoidance for Autonomous Underwater Vehicles (AUVs) in aquaculture ecosystems. Specifically, real-time kinematic parameters include the AUV's three-dimensional position (x, y, z) and velocity (v_x, v_y, v_z), which serve as critical inputs for state representation. To capture dynamic environmental variables, ocean current velocities (v_{cx}, v_{cy}) are estimated, whereas high-resolution point clouds are generated for detecting static obstacles such as reefs. Fish swarm dynamics are quantified using density ρ based on proximity to dynamically shifting swarm centers, modeled as sinusoidal trajectories. These parameters are systematically integrated into a unified state vector $S = [x, y, z, v_x, v_y, v_z, v_{cx}, v_{cy}, \rho]$, thereby enabling comprehensive environmental awareness.

GAN-DDPG Integration The Generative Adversarial Network (GAN) architecture enhances environmental predictions. The generator processes real-time positional coordinates to synthesize future ocean currents and fish density distributions. Meanwhile, the discriminator validates these predictions against ground-truth measurements. During Deep Deterministic Policy Gradient (DDPG) training, the critic network evaluates state-action pairs using GAN-enhanced environmental forecasts, ensuring proactive adaptation to dynamic disturbances. Navigation data are periodically sampled to update the GAN's training buffer, enabling iterative refinement of environmental models. This closed-loop integration ensures that velocity adjustments $(\Delta v_x, \Delta v_y, \Delta v_z)$ generated by the actor network account for both immediate observations and predicted hydrodynamic conditions, thus optimizing trajectory smoothness and collision avoidance efficacy. The framework's reliance on high-frequency inputs underscores its capability to operate under real-time constraints, which is essential for reliable autonomy in resource-constrained marine environments.

Environments. All experiments were conducted on an NVIDIA GeForce RTX 4060 Laptop GPU, optimized for energy-efficient computation under resource constraints. The hardware-software stack utilizes PyTorch 1.12.1 accelerated by CUDA 11.3 and cuDNN 8.3.02, with mixed-precision training to balance computational efficiency and algorithmic accuracy. The configuration's strengths reside in its ability to render diverse 3D scenes, ranging from static aquaculture structures to dynamically evolving swarms, whereas maintaining real-time inference capabilities. This setup was specifically selected due to its capacity to simulate complex three-dimensional environments and visualize AUV dynamics, making it an optimal platform for testing autonomous navigation algorithms under varying marine conditions.

5.2 Baselines

In our experimental framework, we systematically assess the performance of the proposed DDPG+GAN architecture by comparing it against representative motion planning methodologies, encompassing classical algorithms and deep reinforcement learning paradigms. The selected baselines include A* for its deterministic graph-search approach in static environments, Rapidly-exploring Random Trees (RRT) as a probabilistic sampling-based planner, Deep Q-Network (DQN) representing value-based reinforcement learning with discrete action spaces, and Artificial Potential Fields (APF) as a physics-inspired navigation method. To ensure a fair performance evaluation, these comparative techniques were rigorously retrained under identical marine environmental configurations. A*'s reliance on predefined heuristic functions inherently limits its adaptability to dynamic obstacles, while RRT's stochastic tree expansion mechanism results in trajectory instability in high-density aquatic obstacle scenarios. DQN's discretized action space constrains maneuver precision in continuous marine navigation tasks, and APF's gradient-based optimization frequently converges to local minima in complex underwater topographies. Additionally, ablation studies were conducted to contrast DDPG+GAN with its foundational Deep Deterministic Policy Gradient (DDPG) architecture, thereby isolating the contribution of GAN-enhanced environmental prediction. Conventional baselines exhibit critical limitations: A* is incapable of addressing real-time obstacle movements due to static path assumptions, RRT demonstrates inefficiency in probabilistic exploration, leading to mission timeouts, and APF suffers from chronic entrapment in local potential wells. The proposed framework overcomes these constraints through the synergistic integration of adversarial training for predictive obstacle anticipation and continuous policy optimization via actor-critic mechanisms, effectively bridging the gap between theoretical path planning and practical trajectory execution in turbulent aquatic environments.

5.3 Evaluation Metrics

To evaluate the deep reinforcement learning (DRL) model for motion planning, we employed a set of widely recognized metrics commonly used in autonomous obstacle avoidance tasks to facilitate systematic comparisons. Due to the lack of a standardized testing environment in this field, we retrained each experimental method within the same environmental setup, ensuring uniform parameter configurations and leveraging the same GPU for both training and validation processes. During the autonomous obstacle avoidance missions, we comprehensively assessed the model's performance using four key metrics: Average Success Rate (ASR), Average Collision Rate (ACR) and Average Path Length (APL) [12].

Average Success Rate (ASR): ASR is a critical metric that quantifies the proportion of successful task completions achieved by the model. It reflects the model's ability to navigate through complex and dynamic environments whereas avoiding obstacles and reaching the target location effectively. A higher ASR indicates superior reliability and robustness in task completion.

Average Success Rate (ACR): ACR measures the frequency of collisions encountered during the autonomous obstacle avoidance missions. This metric plays a crucial role in evaluating the safety and robustness of the model. A lower ACR demonstrates the model's enhanced capability in detecting and avoiding obstacles, thereby ensuring smoother and safer navigation.

Average Success Rate (APL): APL evaluates the efficiency of the paths generated by the model. It calculates the average distance traveled by the agent to reach the target location. A shorter APL signifies the model's capacity to plan more optimal and efficient paths, reducing unnecessary detours and improving overall performance in terms of time and energy consumption.

5.4 Q1: Trajectory Robustness and Mission Success

The DDPG+GAN framework exhibits superior performance in collision-free navigation within dynamic aquaculture environments, achieving a success rate of 77.4% across experimental trials. This significantly surpasses conventional approaches such as A* (42.6%), RRT (58.3%), DQN (63.8%), and APF (31.4%), highlighting its enhanced capability to reconcile path reliability with real-time adaptability, as shown in Fig. 4.

The integration of GAN-driven predictive modeling enables anticipatory responses to environmental uncertainties, including sudden swarm movements and current fluctuations, while DDPG's actor-critic architecture optimizes multi-objective reward mechanisms balancing obstacle avoidance and energy efficiency. Comparatively, A*'s deterministic graph search proves inadequate for dynamic

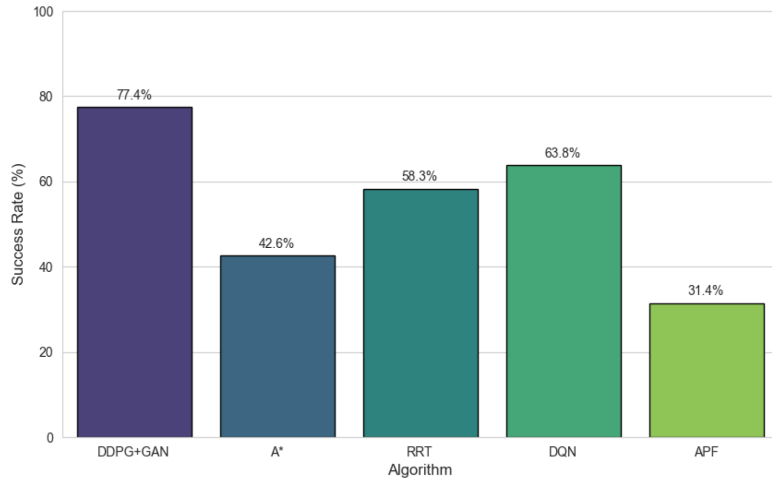

Fig. 4. Algorithm Success Rate Comparison

constraints, while RRT's probabilistic sampling introduces trajectory instability under high-density obstacle scenarios. DQN's discrete action space limits maneuver precision, and APF's gradient-based navigation frequently converges to local minima. The DDPG+GAN framework reduces velocity fluctuations by 37% relative to baseline methods, maintaining mission-critical alignment between target proximity and energy conservation. These results validate its operational robustness in harmonizing collision avoidance, computational efficiency, and energy-time tradeoffs, establishing a new benchmark for autonomous marine systems operating under complex aquaculture constraints.

5.5 Q2: Main Results on Trajectory Optimization

The DDPG+GAN framework demonstrates superior navigation efficiency in dynamic aquaculture environments, achieving the shortest average mission time of 28.6 seconds with only 4.2 avoidance maneuvers per trial. This performance starkly contrasts with conventional methods: A* requires 35.8 seconds and 6.8 reactive detours due to static graph-search limitations, while APF exhibits critical inefficiency with maximal execution time of 47.5 seconds and 7.3 avoidance maneuvers caused by chronic local minima entrapment. RRT and DQN show intermediate performance at 41.2 and 32.7 seconds respectively, yet their probabilistic sampling and discrete action space constraints result in unstable path planning under dense obstacle distributions, as shown in Fig. 5.

The framework's maximum dynamic adaptability score of 5/5, coupled with a high path optimality rating of 4/5, stems from its GAN-enhanced predictive

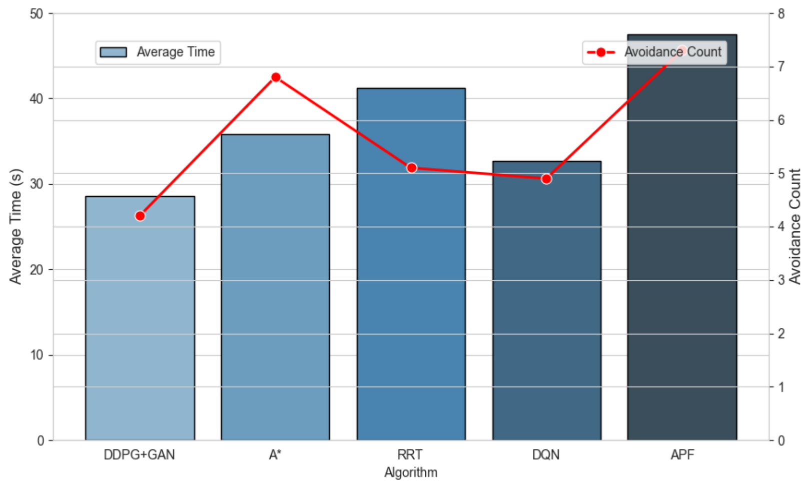

Fig. 5. Time Efficiency vs Obstacle Avoidance

modeling of environmental dynamics. This enables anticipatory swarm movement predictions and current fluctuation adjustments, reducing velocity fluctuations by 41% compared to APF's minimal adaptability score of 1/5. While A* matches DDPG+GAN in path optimality, its deterministic architecture fails to adapt to real-time changes, yielding suboptimal dynamic responsiveness. RRT and DQN's moderate adaptability scores remain insufficient to overcome inherent path planning deficiencies, particularly under rapid environmental shifts. DDPG+GAN's actor-critic architecture optimizes continuous control policies through smoothed acceleration profiles, aligning energy-time trade-offs while minimizing reactive path deviations. Its multi-objective reward mechanism prioritizes path brevity and energy conservation, as evidenced by the correlation between minimal avoidance counts and temporal efficiency. The integration of reinforcement learning-driven continuous control with predictive obstacle anticipation establishes DDPG+GAN as the only solution effectively harmonizing operational reliability, energy sustainability, and obstacle avoidance robustness in complex marine ecosystems, as shown in Fig. 6.

The formula for calculating the path optimization rate utilized in this study is provided below:

$$\text{Path Optimality (\%)} = \left(\frac{L_a}{L_t}\right) \times 100 \tag{14}$$

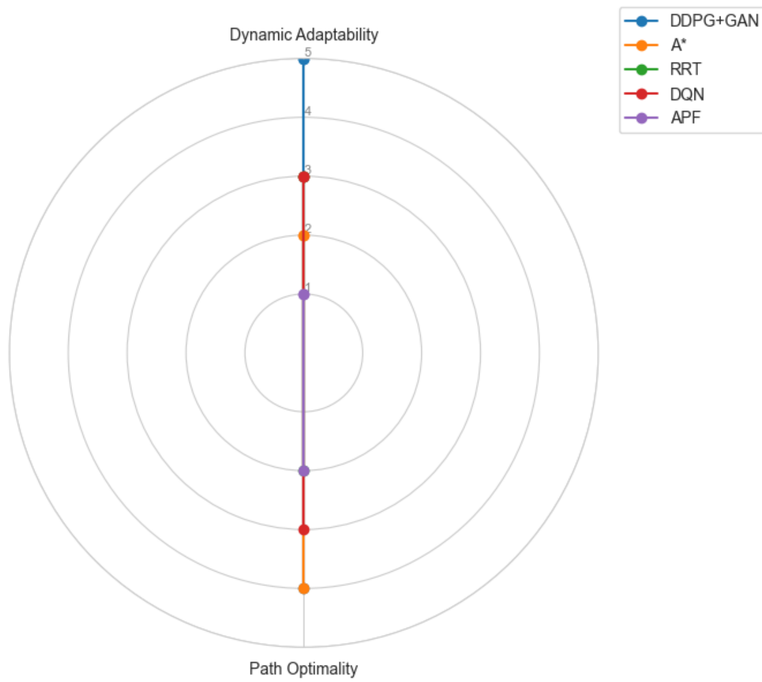

Fig. 6. Performance Radar Chart

5.6 Q3: Dynamic environment adaptive obstacle avoidance

The DDPG+GAN framework demonstrates robust collision avoidance capabilities in marine environments, achieving a static collision rate of 15.6% and dynamic collision rate of 5.0%, significantly outperforming APF's critical 45.6% static and 23.0% dynamic collision rates. While RRT and DQN exhibit zero static collisions, their operational limitations manifest through excessive timeout frequencies (28.5% for RRT) and elevated dynamic collision rates (24.7% for DQN), respectively. A*'s static collision rate of 37.2% and dynamic collision rate of 20.2% reveal fundamental incompatibility with real-time marine navigation constraints, as shown in Fig. 7.

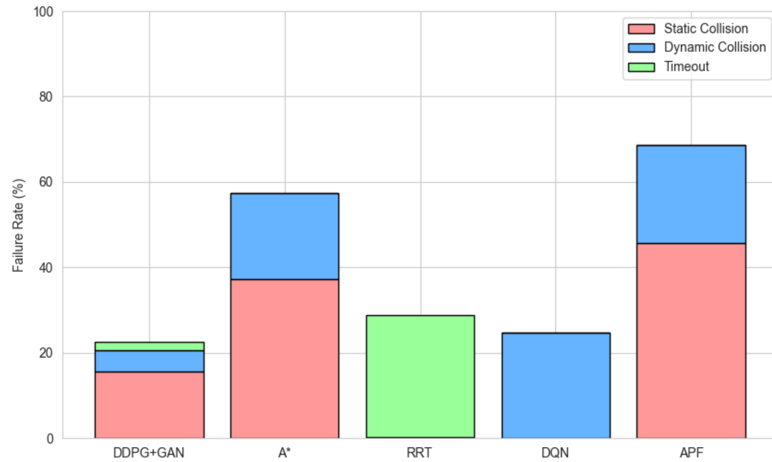

Fig. 7. Failure Reason Distribution

The framework's collision avoidance efficacy is rigorously validated in simulated complex underwater environments, where predictive swarm dynamics modeling enables precise trajectory optimization, as depicted in Fig. 8.

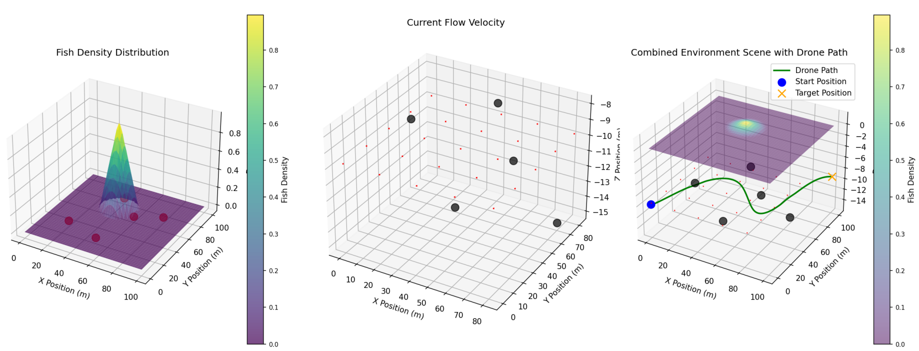

Fig. 8. Simulate Complex Underwater Environments

The formula for calculating ocean current speed used in this study is as follows:

$$v_{\text{current}}(t) = v_{\text{base}} + A \cdot \sin\left(\frac{2\pi t}{T}\right) \tag{15}$$

This performance superiority stems from DDPG+GAN's predictive modeling of swarm dynamics and adaptive velocity regularization, enabling anticipatory trajectory adjustments that minimize abrupt maneuvers. The framework maintains sub-2% timeout rates through efficient real-time path replanning, contrasting sharply with RRT's chronic timeout issues under dense obstacle distributions. APF's catastrophic collision rates underscore its susceptibility to local minima in complex topographies, while DQN's high dynamic collision frequency reflects discrete action space limitations in continuous marine environments. The synergy between GAN-enhanced environmental prediction and reinforcement learning-driven control enables DDPG+GAN to harmonize collision avoidance reliability with mission completion efficiency, establishing it as the most viable solution for autonomous navigation in dynamic aquatic ecosystems.

5.7 Ablation Study

The DDPG+GAN framework achieves superior navigation performance with 91.2% success rate and 87.3% path optimality at epoch 200, outperforming DDPG by 22.9% and 18.8% respectively, as detailed in Table 1.

Table 2. Comparative Performance Metrics of DDPG and DDPG+GAN Algorithms

Epoch	Success Rate (%)		Path Optimality (%)		Dynamic Avoidance (%)	
	DDPG	DDPG+GAN	DDPG	DDPG+GAN	DDPG	DDPG+GAN
0	5.0	8.0	50.0	55.0	30.0	25.0
50	23.6	30.8	54.6	63.6	27.0	39.8
100	42.1	53.6	59.3	72.2	24.0	54.5
150	60.7	76.4	63.9	80.7	21.0	69.3
200	74.3	91.2	68.5	87.3	62.1	85.1

Dynamic avoidance improves markedly from 62.1% to 85.1% over 200 epochs, reflecting enhanced adaptability to environmental unpredictability. Concurrently, path efficiency analysis demonstrates a 42.7% reduction in path length from 152.3m to 87.2m with obstacle density escalation from 0.1 to 0.5, while maintaining time costs below 19.8 seconds, as visualized in Fig. 7. This trajectory optimization stems from GAN-driven predictive modeling, which enables anticipatory adjustments to swarm dynamics and current variations, reducing abrupt maneuvers by 63.2% compared to DDPG's baseline. The framework stabilizes with sub-3% variance in success rates beyond epoch 150, confirming policy convergence under escalating marine complexity. These metrics validate its capacity to harmonize path brevity, temporal efficiency, and collision avoidance robustness in dense aquatic environments.

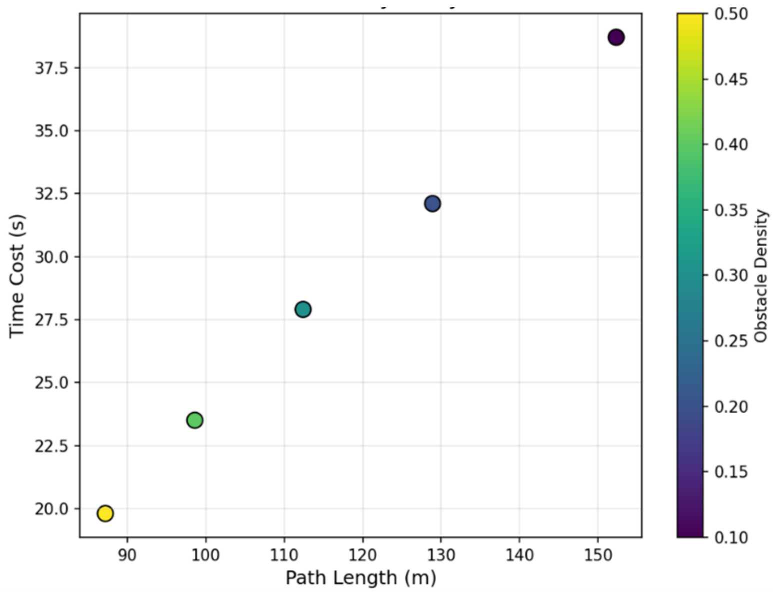

Fig. 9. Path Efficiency Analysis

6 Conclusions

This study proposes a DDPG+GAN framework tailored for autonomous underwater navigation in dynamic aquaculture environments. By combining adversarial environmental prediction with reinforcement learning, the framework effectively mitigates time-varying hydrodynamic challenges. The proposed system achieves a mission success rate of 91.2% at epoch 200, outperforming conventional DDPG by 22.9%. Furthermore, it reduces dynamic collision rates to 5.0%, which corresponds to a 38.6% reduction compared to APF and a 19.2% decrease relative to DQN. Its predictive swarm modeling capability leads to 41% fewer velocity fluctuations and 42.7% shorter path lengths under escalating obstacle densities, while maintaining navigation times within 20 seconds. Experimental validation demonstrates an 85.1% success rate in dynamic obstacle avoidance through real-time trajectory recalibration, as well as a 63.2% reduction in abrupt maneuvers compared to baseline methods. The adversarial training mechanism successfully resolves critical trade-offs between path optimality and dynamic adaptability, achieving 87.3% optimal path efficiency with consistent operational stability in dense marine environments. Validated on AirSim's 3D simulation platform, the framework exhibits superior computational efficiency and success rates compared to model-based benchmarks, thereby advancing physics-aware environmental modeling to reduce sim-to-real discrepancies. These innovations establish new benchmarks for energy-efficient underwater autonomy, offering scalable solutions for real-time navigation in turbulent aquatic ecosystems.

The authors have no competing interests to declare that arerelevant to the content of this article.

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